Junsheng Huang

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github.com/no-touch-fish

Education

Zhejiang University-University of Illinois at Urbana-Champaign Aug. 2021 - Jun. 2025

Dual Bachelor of Electrical and Computer Engineering (UIUC GPA: 4.00 / 4.00) (ZJU GPA: 3.98 / 4.00)

- **Relevant Courses:** Computer System Engineering (ECE391), Data Structure (CS225), Artificial Intelligence (CS440), Algorithm (CS374), Machine Learning (CS446), Applied Parallel Programming (ECE408)

University of Illinois at Urbana-Champaign Aug. 2025 – May 2027 (expected)

Master of Science in Electrical and Computer Engineering

- **Relevant Courses:** Reinforcement Learning (ECE586), Natural Language Processing (CS546), Computer Vision (CS543), Graph Machine Learning (CS598)

Research Interests

LLM Agents, LLM Post-training, Trustworthy LLMs, AI for Engineering Design

Publications

- **Junsheng Huang**, Yifan Sun, Zhouer Zhang, Ning Wei, Yening Liu, Michael Molter, Pavan Kumar Hanumolu, Elyse Rosenbaum, Bin Hu, Huan Zhang. *SAFTAC: Simulation-Augmented Fine-Tuning of Open-Source LLMs for Analog Circuit Design*. Manuscript under review.
- **Junsheng Huang**, Zhitao He, Sandeep Polisetty, Yi R. (May) Fung. *MAC-Tuning: LLM Multi-Compositional Problem Reasoning with Enhanced Knowledge Boundary Awareness*. Accepted to **EMNLP** 2025 Main Conference.
- Jason Vega, **Junsheng Huang**, Gaokai Zhang, Hangoo Kang, Minjia Zhang, Gagandeep Singh. *Stochastic Monkeys at Play: Random Augmentations Cheaply Break LLM Safety Alignment*

Research Experience

SAFTAC: Simulation-Augmented Fine-Tuning of Open-Source LLMs for Analog Circuit Design | Advisor: *Huan Zhang*

- Led a project on post-training of open-source LLMs for specification-conditioned analog circuit design.
- Created an automated pipeline for generating and annotating analog design data using LLM assistance, producing a comprehensive benchmark of 8,626 tasks across 10 circuit types
- Designed a multi-stage post-training framework comprising a three-stage supervised fine-tuning curriculum and subsequent GRPO-based reinforcement learning.
- Fine-tuned Qwen3-8B, achieving superior performance over closed-source LLMs including Claude Opus 4.7 and GPT-5.4, along with leading open-source LLMs.

MAC-Tuning: LLM Multi-Compositional Problem Reasoning with Enhanced Knowledge Boundary Awareness
Advisor: *Yi R. (May) Fung*

- Led a project on improving LLM hallucination detection and reasoning under multi-problem compositional settings.
- Proposed MAC-Tuning, a stepwise fine-tuning framework that builds multi-problem QA and confidence data to decouple answer prediction from confidence estimation, improving knowledge-boundary awareness.
- Built an automatic evaluation framework with LLM-as-a-judge and keyword-based extraction, assessing answer correctness and confidence calibration via multiple metrics.

Random Augmentations Cheaply Break LLM Safety Alignment | Advisor: *Gagandeep Singh*

- Contributed to a project studying how simple random prompt augmentations affect LLM safety alignment.
- Implemented multiple low-cost input augmentation strategies, including different level of perturbation.
- Co-designed safety evaluation metrics and manually annotated model outputs to compare LLM-based classifiers with human judgments, revealing that simple random augmentations can significantly weaken safety-aligned LLMs.

Projects

Strategic Persuasion in LLMs: Multi-Turn Dialogue with Reinforcement Learning

- Developed a planner-persuader LLM agent for multi-turn strategic persuasion, fine-tuning Qwen3-4B with GRPO and designing composite rewards for strategy following, format control, and attitude-shift improvement.

Federated Knowledge-Augmented Generation

- Built a federated KAG/RAG pipeline based on LightRAG for privacy-preserving personalized text generation, covering data collection, codebase construction, and core training and generation workflows.

Language-Guided Robotic Grasping System

- Built an embodied AI agent that maps spoken instructions to robotic grasping actions through ASR, NLP-based target extraction, YOLO-based object detection, 6D pose estimation, and motion planning.

Technical Skills

Programming Languages: Python, C/C++, CUDA, x86 Assembly, MATLAB
Frameworks & Tools: PyTorch, Hugging Face Transformers, PEFT/LoRA, DeepSpeed/FSDP, vLLM, verl
Spoken Languages: English (Proficient), Mandarin (Native), Cantonese (Native)

Honors

Honorable Mention of Mathematical Contest of Modeling (**May 2023**)

ZJU-UIUC Institute Dean's List, **Sophomore Year**

ZJU-UIUC Institute Third-Class Academic Excellence Award, **Sophomore and Junior Years**

UIUC Grainger Engineering Department Dean's List for two semesters, **Junior Year**

Teaching and Leadership

Teaching Assistant for CS446/ECE449 (Machine Learning)	Jan 2026 - May 2026
Teaching Assistant for MATH285 (Introduction to Differential Equations)	Jan 2025 - Jun 2025
Teaching Assistant for MATH241 (Calculus III)	Sep 2024 - Dec 2024
Course Assistant for ECE391 (Computer System Engineering)	Jan 2024 - May 2024
Student Representative for ECE Major in ZJUI	Sep 2021 - Jun 2022, Sep 2024 - Jun 2025